

# MIAE 221: Materials Science for Engineers

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## Chapter 6: Mechanical Properties of Metals (Expanded & Condensed Guide)

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Concordia University · Gina Cody School of Engineering

**Textbook:**

*Materials Science and Engineering: An Introduction*

(10th Ed., Callister & Rethwisch)

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*(Syllabus Week 6 · Core Midterm & Laboratory Topic)*

### 1. 🌀 30-Second Core Intuition & Plain-English Concept

When you pull a metal rod in tension, it initially stretches like a stiff rubber band: release the load, and it snaps back to its original length (**Elastic Deformation**).

Pull it past its **Yield Strength** ( $\sigma_y$ ), and atomic planes permanently slide past one another via dislocation slip (**Plastic Deformation**). At the peak load (**Ultimate Tensile Strength, UTS**), the cross-section necks down until fracture occurs.

*Key Vocabulary Distinction*

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- **Stiffness** ( $E$ ): Resistance to elastic stretching (slope of elastic region).
- **Strength** ( $\sigma_y$ , UTS): Resistance to permanent plastic deformation and tearing.
- **Ductility** ( $\epsilon$ ): Total plastic deformation sustained before snap.
- **Toughness**: Total energy absorbed up to fracture (area under entire curve).

## 2. ⚙️ High-Yield Mathematical Engine & Tensile Metrics

| Property                      | Formula                                                                    | Description & Units                                                                                         |
|-------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Engineering Stress            | $\sigma = \frac{F}{A_0}$                                                   | Force divided by<br><i>original</i><br>cross-sectional area ( <i>extMPa</i> or <i>extN/mm<sup>2</sup></i> ) |
| Engineering Strain            | $\epsilon = \frac{\Delta l}{l_0} = \frac{l_i - l_0}{l_0}$                  | Elongation divided by<br><i>original</i><br>gauge length (dimensionless or <i>extmm/mm</i> )                |
| Hooke's Law                   | $\sigma = E\epsilon$                                                       | Linear elastic response ( <i>E</i> : Young's modulus, <i>extGPa</i> )                                       |
| Poisson's Ratio               | $\nu = -\frac{\epsilon_x}{\epsilon_z} = -\frac{\epsilon_y}{\epsilon_z}$    | Lateral contraction divided by longitudinal expansion (typically $\approx 0.33$ )                           |
| Yield Strength ( $\sigma_y$ ) | Stress at 0.002 (0.2) offset                                               | Parallel line with slope <i>E</i> drawn starting at $\epsilon = 0.002$                                      |
| Ductility (%EL)               |                                                                            | Total permanent elongation at fracture                                                                      |
| Modulus of Resilience         | $U_r = \int_0^{\epsilon_y} \sigma d\epsilon \approx \frac{\sigma_y^2}{2E}$ | Elastic energy absorbed per unit volume without permanent deformation                                       |
| True Stress ( $\sigma_T$ )    | $\sigma_T = \frac{F}{A_i} = \sigma(1 + \epsilon)$                          | Instantaneous force / instantaneous area (valid up to necking)                                              |
| True Strain ( $\epsilon_T$ )  | $\epsilon_T = \ln\left(\frac{l_i}{l_0}\right) = \ln(1 + \epsilon)$         | Logarithmic true strain (valid up to necking)                                                               |

### 3. Textbook Reference Diagram & Visual Pedagogical Analysis

yield point; stress subsequently rises with increasing strain. For metals that display this effect, the yield strength is taken as the average stress that is associated with the lower yield point because it is well defined and relatively insensitive to the testing procedure.<sup>10</sup> Thus, it is not necessary to employ the strain offset method for these materials.

The magnitude of the yield strength for a metal is a measure of its resistance to plastic deformation. Yield strengths may range from 35 MPa (5000 psi) for a low-strength aluminum to greater than 1400 MPa (200,000 psi) for high-strength steels.

**Concept Check 6.1** Cite the primary differences between elastic, anelastic, and plastic deformation behaviors.

The answer may be found in all digital versions of the text and/or at [www.wiley.com/college/callister](http://www.wiley.com/college/callister) (Student Companion Site).]

#### Tensile Strength

After yielding, the stress necessary to continue plastic deformation in metals increases to a maximum, point *M* in Figure 6.11, and then decreases to the eventual fracture, point *F*. The **tensile strength** *TS* (MPa or psi) is the stress at the maximum on the engineering stress-strain curve (Figure 6.11). This corresponds to the maximum stress that can be sustained by a structure in tension; if this stress is applied and maintained, fracture will result. All deformation to this point is uniform throughout the narrow region of the tensile specimen. However, at this maximum stress, a small constriction or neck begins to form at some point, and all subsequent deformation is confined at this neck, as indicated by the schematic specimen insets in Figure 6.11. This phenomenon is termed *necking*,

in lieu of *stress* because strength is a property of the metal, whereas stress is related to the magnitude of the force. In the United States, the unit of kilopounds per square inch (ksi) is sometimes used for the sake of convenience, where

Figure 6.11: Engineering stress-strain behavior to fracture: *P* (proportional limit), *M* (maximum load / tensile strength), and *F* (fracture point).

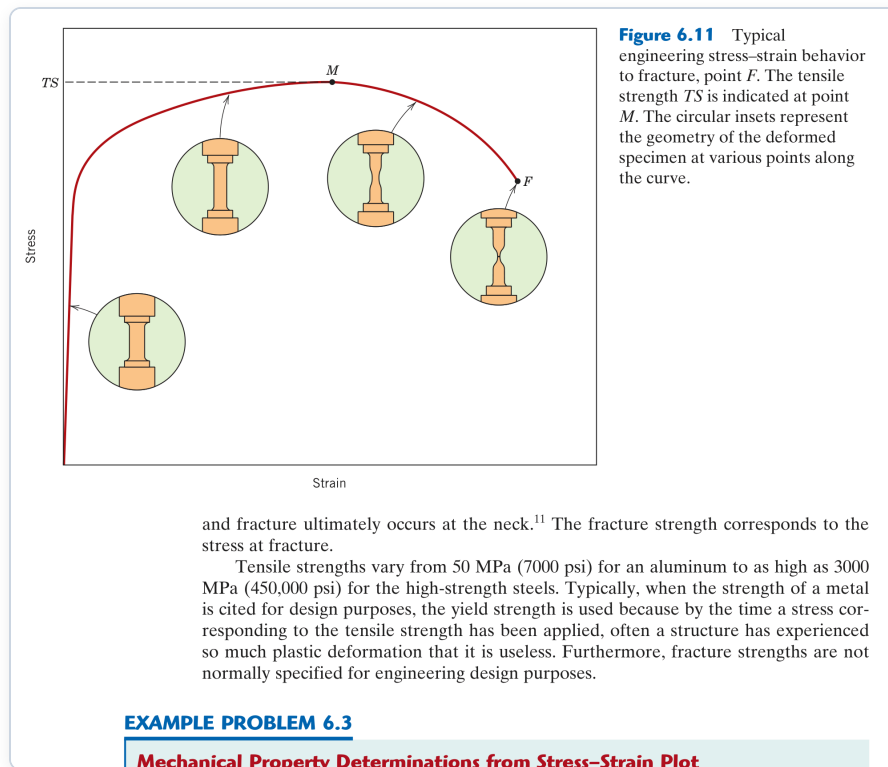


Figure 6.12: Determination of the 0.002 (0.2) strain offset yield strength for brass.

## In-Depth Visual Breakdown:

- **The Engineering Curve (Figure 6.11):**

- The curve appears to drop after peak  $M$  (Tensile Strength) because engineering stress uses *original area*

$A_0$ . In reality, the material continues to strain harden, but localized **necking** causes the cross-section to shrink faster than the load increases.

- **The 0.002 Offset Method (Figure 6.12):**

- For metals without a sharp yield drop (like brass, aluminum, copper), yield strength is standardized by starting at  $\epsilon = 0.002$  (0.2) on the horizontal axis and drawing a dashed line strictly **parallel** to the initial linear elastic slope  $E$ . The intersection with the experimental curve defines  $\sigma_y$ .

## 4. 🏆 Teacher Notes Cross-Reference & Concordia Exam Traps

- **Concordia Exam & Lab Focus:**

- Calculating Young's modulus, yield strength (0.2), UTS, and ductility directly from raw load-elongation tensile data.
- Modulus of resilience calculation:  $U_r = \frac{\sigma_y^2}{2E}$ .
- Hardness correlations: For steel,  $\text{UTS (MPa)} \approx 3.45 \times \text{HB}$  (Brinell Hardness).

- **Concordia Exam Traps:**

- **True vs. Engineering Equations:** The equations  $\sigma_T = \sigma(1 + \epsilon)$  and  $\epsilon_T = \ln(1 + \epsilon)$  are **ONLY valid up to the onset of necking (UTS)!** After necking initiates, cross-sectional deformation is no longer uniform, and instantaneous cross-sectional area must be measured directly.

## 5. 💡 Master Exam Problem & Step-by-Step Solution Framework

### Problem:

A cylindrical specimen of aluminum alloy with diameter  $d_0 = 12.8 \text{ mm}$  and gage length  $l_0 = 50.8 \text{ mm}$  is loaded in tension. An elastic load of  $15,000 \text{ N}$  produces an elongation of  $0.089 \text{ mm}$ . The specimen yields at  $30,000 \text{ N}$ , reaches a maximum load of  $45,000 \text{ N}$ , and fractures at  $l_f = 56.4 \text{ mm}$ . Calculate: (a) Young's Modulus  $E$ , (b) Yield Strength  $\sigma_y$ , (c) Tensile Strength UTS, (d) Ductility, and (e) Modulus of Resilience  $U_r$ .

- **Step 1: Compute Initial Area  $A_0$**

$$A_0 = \frac{\pi d_0^2}{4} = \frac{\pi (12.8 \text{ mm})^2}{4} = 128.7 \text{ mm}^2 = 1.287 \times 10^{-4} \text{ m}^2$$

- **Step 2: Compute Modulus of Elasticity  $E$**

$$\sigma_1 = \frac{15,000 \text{ N}}{128.7 \text{ mm}^2} = 116.55 \text{ MPa}, \quad \epsilon_1 = \frac{0.089 \text{ mm}}{50.8 \text{ mm}} = 0.001752$$
$$E = \frac{\sigma_1}{\epsilon_1} = \frac{116.55 \text{ MPa}}{0.001752} = 66,524 \text{ MPa} = 66.5 \text{ GPa}$$

- **Step 3: Compute Yield Strength  $\sigma_y$  and Tensile Strength UTS**

$$\sigma_y = \frac{F_{\text{yield}}}{A_0} = \frac{30,000 \text{ N}}{128.7 \text{ mm}^2} = 233.1 \text{ MPa}$$

$$\text{UTS} = \frac{F_{\max}}{A_0} = \frac{45,000 \text{ N}}{128.7 \text{ mm}^2} = 349.6 \text{ MPa}$$

- **Step 4: Compute Ductility**

- **Step 5: Compute Modulus of Resilience  $U_r$**

$$U_r \approx \frac{\sigma_y^2}{2E} = \frac{(233.1 \times 10^6 \text{ Pa})^2}{2(66.5 \times 10^9 \text{ Pa})} = 4.08 \times 10^5 \text{ J/m}^3 = 0.408 \text{ MJ/m}^3$$


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